

Ephaptic A β -to-C-Fiber Crosstalk and Multi-Fiber Spatial Summation: A Closed-Loop Core-Conductor Study with Nociceptor-Realistic Channel Kinetics

Mina Saied Attia Rizk

New Cairo STEM School, Cairo, Egypt

Corresponding author: Mina.3024031@stemnewcairo.moe.edu.eg

Abstract. Ephaptic excitation of unmyelinated C-fiber nociceptors by myelinated A β afferents at sites of nerve injury has been proposed as a mechanism of tactile allodynia. We tested whether it can reach threshold in a closed-loop core-conductor model. A CRRSS A β axon and a C-fiber with Hodgkin–Huxley or phenomenological Nav1.8/Nav1.9 kinetics share a restricted extracellular compartment along a 5-mm lesion. Intracellular and extracellular potentials are solved together, and every endpoint is shown to converge. A single A β action potential depolarized the C-fiber by 1.56 mV. Synchronous volleys, equivalent to smaller extracellular cross-sections per fiber, raised this to at most 26.9 mV; the response was limited not by saturation but by failure of A β conduction inside the confined lesion. The depolarization lasted ~ 70 μ s, was spatially narrow and was flanked by hyperpolarization, so no fixed voltage threshold applied: a 0.1-ms pulse could drive the Nav1.8/Nav1.9 membrane to 39 mV without firing it, and the recorded ephaptic drive had to be amplified 4.6-fold to fire it at 25 synchronous fibers. No C-fiber action potential occurred over five decades of extracellular leak, lesions of 1–7 mm, stimuli of 1.2–20 $\times$ threshold, 50–400-Hz trains or sensitizing bias currents, and a 0.5-ms onset dispersion removed 85 % of the drive. Faster gating alone had no effect; action potentials appeared only when the whole Hodgkin–Huxley membrane was made faster (time constant ≤ 0.37 ms), never with Nav1.8/Nav1.9 kinetics, whose activation lies 42 mV above rest. Direct ephaptic A β $\rightarrow$ C excitation therefore requires conditions that also block the A β source.

Keywords: ephaptic coupling · core-conductor model · C-fiber nociceptor · Nav1.8 · mechanical allodynia · numerical convergence

1 Introduction

Dynamic mechanical allodynia, pain evoked by light moving touch, is a hallmark of many peripheral neuropathic conditions. One peripheral mechanism proposed for it is ephaptic coupling: at sites of demyelination or compression, current generated by large myelinated A β afferents might flow through a shared, restricted extracellular space and excite neighbouring unmyelinated C-fiber nociceptors directly. Ephaptic transmission between single fibers has been demonstrated in injured or dysmyelinated preparations: in chronically damaged peripheral nerves (Seltzer & Devor, 1979), in the amyelinated spinal roots of dystrophic mice (Rasminsky, 1980) and among the densely packed unmyelinated axons of the olfactory nerve (Bokil et al., 2001). Even between intact myelinated afferents in rat dorsal roots, passing impulses transiently raise the excitability of neighbouring fibers (Bolzoni & Jankowska, 2019). Rapid ephaptic interactions were first characterized between adjacent fibers by Katz and Schmitt (1940) and Arvanitaki (1942). By contrast, the cross-excitation between A- and C-neurons observed in dorsal root ganglia (Devor & Wall, 1990; Amir & Devor, 2000) is slow and has been attributed largely to non-synaptic chemical mediation rather than to ephaptic current flow (Amir & Devor, 1996). It is therefore not direct evidence for ephaptic A β $\rightarrow$ C excitation in nerve trunks. Whether ephaptic A β $\rightarrow$ C coupling can reach the C-fiber's firing threshold is a quantitative question that is hard to settle experimentally, which motivates modelling.

Computational studies of ephaptic coupling either impose the field of an active fiber on a passive or excitable target without feedback, as in the cable treatment of endogenous fields by Goldwyn and Rinzel (2016), or solve the coupled

intracellular–extracellular problem for bundles of similar fibers. The latter include closed-loop core-conductor models of myelinated or demyelinated bundles (Binczak et al., 2001; Reutskiy et al., 2003; Schmidt et al., 2021; Schmidt & Knösche, 2022), sheets of coupled axons (Sheheitli & Jirsa, 2020), higher-dimensional formulations of interacting tracts (Chawla et al., 2021; Chawla & Morgera, 2024), resistor-network models of whole nerve trunks (Capllonch-Juan & Sepulveda, 2020), and fully resolved extracellular–membrane–intracellular (EMI) models (Jæger & Tveito, 2026). The coupling between two physiologically dissimilar fiber classes, a fast saltatory A β fiber and a slow, thin C-fiber, has received less attention. That pairing raises its own questions: whether a brief A β transient can excite a C-fiber whose membrane is slow, whether synchronous volleys of many A β fibers can make up for weak single-pair coupling, and how the answer depends on unmeasured extracellular parameters.

Here we address these questions with a closed-loop core-conductor model of a CRRSS A β fiber and a C-fiber that share a restricted extracellular compartment along a focal lesion. The C-fiber has either classical Hodgkin–Huxley kinetics (Phase 1) or a phenomenological Nav1.8/Nav1.9 membrane (Phase 2), within otherwise identical models. An earlier version of this work reported saturating multi-fiber summation and a fixed Nav1.8 voltage threshold. Both findings arose from a lagged coupling scheme whose time-step error grows with the number of fibers, and from a stimulation site placed inside the extracellular compartment. We therefore solve the coupled system monolithically and verify convergence for every endpoint. We place the A β stimulus outside the lesion, characterize the C-fiber’s excitability for brief inputs directly rather than through a nominal voltage threshold, and map the outcome over the number of synchronous fibers (equivalently, the extracellular cross-section per fiber), the extracellular leak, the lesion length, stimulus strength, temporal dispersion, repetitive trains, sensitizing bias currents and the speed of C-fiber kinetics.

2 Methods

2.1 Geometry and governing equations

We model one myelinated A β axon type and one unmyelinated C-fiber, both of length $L = 10$ mm, that share a restricted extracellular compartment along a focal lesion ($z = 2.5$ – 7.5 mm; Fig. 1a). Outside the lesion the fibers lie in grounded bulk fluid ($u_e = 0$). The A β action potential (AP) is evoked at node 0 ($z = 0$), outside the lesion, by a 0.2-ms intracellular current pulse of twice the threshold of the uncoupled fiber, and enters the lesion as a propagating wave. n identical A β fibers are stimulated synchronously (Section 2.5); a single C-fiber is the test fiber.

For fiber k ($1 = \text{A}\beta$, $2 = \text{C}$) with intracellular potential u_k , extracellular potential u_e and transmembrane potential $v_k = u_k - u_e$, the core-conductor equations are (Fig. 1b)

$$\pi d_k (C_{m,k} \frac{\partial v_k}{\partial t} + I_{\text{ion},k}) = g_k \frac{\partial^2 (v_k + u_e)}{\partial z^2} + \pi d_k I_{\text{stim},k}, \quad g_k = \frac{1}{r_{i,k}} = \frac{\pi d_k^2}{4\rho_i},$$

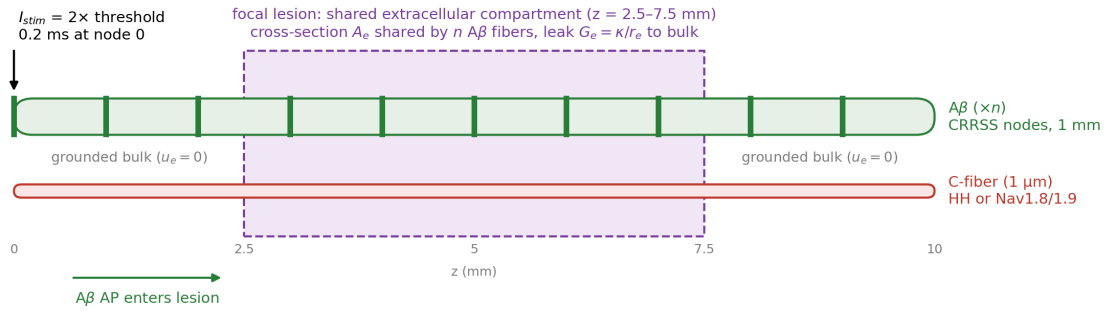
where d_k is the axoplasmic diameter (7 μm for A β , i.e. an outer diameter of 10 μm and a g-ratio of 0.7; 1 μm for the C-fiber), ρ_i the axoplasmic resistivity, and $C_{m,k}$, $I_{\text{ion},k}$, $I_{\text{stim},k}$ are per unit membrane area. Inside the lesion, conservation of current in the shared compartment gives

$$\frac{1}{r_e} \frac{\partial^2 u_e}{\partial z^2} - G_e u_e + n i_{m,1} + i_{m,2} = 0, \quad i_{m,k} = g_k \frac{\partial^2 (v_k + u_e)}{\partial z^2},$$

with $r_e = \rho_e/A_e$ the longitudinal resistance per unit length of a compartment of cross-section A_e , and G_e a transverse leak conductance per unit length from the compartment to grounded bulk tissue. We parameterize the leak by $\kappa = r_e G_e$ (m^{-2}), so that $\lambda_e = \kappa^{-1/2}$ is the passive length constant of the compartment alone. Outside the lesion $u_e = 0$. The

cable ends are sealed. Any current injected by the stimulating electrode leaves through the membrane of the stimulated compartment, which lies outside the lesion in all production runs.

a



b

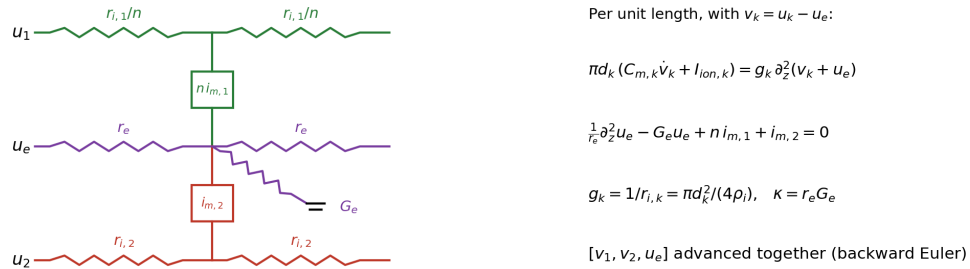


Fig. 1 Model. (a) Geometry: n synchronously stimulated A β fibers (CRRSS nodes every 1 mm) and one C-fiber share an extracellular compartment of cross-section A_e along a 5-mm focal lesion; outside the lesion the fibers lie in grounded bulk fluid. The A β stimulus ($2\times$ threshold, 0.2 ms) is applied at node 0, outside the lesion. (b) Equivalent circuit per unit length and governing equations; the three fields are advanced together.

2.2 Membrane models

A β fiber. Nodes of Ranvier (length $l_{\text{node}} = 1 \mu\text{m}$, spacing 1 mm, ten nodes at $z = 0, 1, \dots, 9 \text{ mm}$) carry CRRSS kinetics (Chiu et al., 1979; Sweeney et al., 1987):

$$\alpha_m = \frac{97 + 0.363 Y}{1 + e^{(31-Y)/5.3}}, \quad \beta_m = \frac{\alpha_m}{e^{(Y-23.8)/4.17}}, \quad \beta_h = \frac{15.6}{1 + e^{(24-Y)/10}}, \quad \alpha_h = \frac{\beta_h}{e^{(Y-5.5)/5}} \text{ (ms}^{-1}\text{)},$$

with $Y = V + 80 \text{ mV}$ and $I_{\text{Na}} = \bar{g}_{\text{Na}} m^2 h (V - E_{\text{Na}})$. The rate constants are used as given by Sweeney et al. (1987) for 37°C , without further temperature scaling. Internodes are passive compact myelin. A computational compartment of width Δz containing a node is homogenized with node fraction $f = l_{\text{node}}/\Delta z$, so that nodal membrane area, capacitance and conductances are independent of the grid (Table 1). Two values depart from Sweeney et al. (1987) and are modelling choices: the nodal capacitance (2.0 rather than $2.5 \mu\text{F cm}^{-2}$) and the g -ratio (0.7 rather than 0.6).

C-fiber, Phase 1 (HH). Classical Hodgkin–Huxley kinetics (Hodgkin & Huxley, 1952) with $\bar{g}_{\text{Na}} = 120$, $\bar{g}_{\text{K}} = 36$, $g_{\text{L}} = 0.3 \text{ mS cm}^{-2}$, $E_{\text{Na}} = +50$, $E_{\text{K}} = -77$, $E_{\text{L}} = -54.4 \text{ mV}$ (resting potential -65.00 mV). The rate functions are the squid-axon rates at 6.3°C , multiplied by $\phi = 3^{(T-6.3)/10}$ when a temperature T is specified (Section 2.6).

C-fiber, Phase 2 (NavC). A phenomenological, calibrated TTX-resistant membrane: Nav1.8 ($m_8^3 h_8$), persistent Nav1.9 (m_9), the HH delayed rectifier (n^4) and a leak,

$$I_{\text{ion},2} = \bar{g}_{18} m_8^3 h_8 (V - E_{\text{Na}}) + \bar{g}_{19} m_9 (V - E_{\text{Na}}) + \bar{g}_{\text{K}} n^4 (V - E_{\text{K}}) + g_{\text{L}} (V - E_{\text{L}}),$$

with Boltzmann steady states ($V_{1/2}, k$) = (−25, 6), (−42, 6) and (−50, 5) mV for m_8 , h_8 (inactivating) and m_9 , and voltage-independent time constants $\tau_{m8} = 1.5$ ms, $\tau_{h8} = 2$ ms and $\tau_{m9} = 10$ ms. The activation parameters are in the range reported for TTX-resistant currents of small DRG neurons (Akopian et al., 1996; Cummins et al., 1999), but the model is a simplification: the time constants are voltage independent, there is no Nav1.7 and only one K^+ current, and two parameters were calibrated rather than taken from data. With the literature-range inactivation values ($V_{1/2} \approx -30$ mV, $\tau_{h8} \approx 17$ ms) the reduced membrane settled on a non-repolarizing depolarized state, so $V_{1/2}$ was moved to −42 mV and τ_{h8} to 2 ms; $\bar{g}_K$ was raised to 60 mS cm^{−2}, and $\bar{g}_{18} = 2000$ and $\bar{g}_{19} = 0.2$ mS cm^{−2} were chosen to give an overshooting AP with APD₅₀ $\approx$ 7 ms. Resting potential −66.82 mV.

Shared cable properties. Both C-fiber models use the same cable (diameter 1 μ m, $C_m = 1$ μ F cm^{−2}, $\rho_i = 54.7$ Ω cm), and ρ_i is the same for the A β fiber. Phase 1 and Phase 2 therefore differ only in C-fiber membrane kinetics; the A β model, stimulus, geometry, numerics and endpoints are identical. All parameters are collected in Table 1 and are read from a single parameter module in the code.

Group	Quantity	Value	Units	Source / status
A β fiber	outer diameter D_i ; g-ratio; axon diameter d_i	10; 0.7; 7	μ m; −; μ m	g-ratio: modelling choice (Sweeney 1987: 0.6)
A β fiber	node spacing; node length	1; 1	mm; μ m	modelling choice
A β fiber	C_m node; C_m myelin	2; 0.005	μ F cm ^{−2}	node: modelling choice (Sweeney 1987: 2.5)
A β fiber	$\bar{g}_{Na}$; $\bar{g}_{L}$ node; $\bar{g}_{L}$ myelin	1445; 128; 0.006	mS cm ^{−2}	Sweeney et al. 1987; myelin: assumption
A β fiber	E_{Na} ; E_{L}	+35.64; −80.0	mV	Sweeney et al. 1987
A β fiber	CRRSS rates	as published for 37 °C ($\phi = 1$)	–	Chiu et al. 1979; Sweeney et al. 1987
Axoplasm	ρ_i (all fibers)	54.7	Ω cm	Sweeney et al. 1987
C-fiber cable	diameter; C_m	1; 1	μ m; μ F cm ^{−2}	both C-fiber models
HH C-fiber	$\bar{g}_{Na}$; $\bar{g}_K$; $\bar{g}_{L}$	120; 36; 0.3	mS cm ^{−2}	Hodgkin & Huxley 1952
HH C-fiber	E_{Na} ; E_K ; E_{L}	+50; −77; −54.4	mV	Hodgkin & Huxley 1952
HH C-fiber	kinetics temperature	6.3 °C (varied 6.3–25 °C)	–	squid rates; $Q_{10} = 3$ when varied
NavC C-fiber	$\bar{g}_{18}$; $\bar{g}_{19}$; $\bar{g}_K$; $\bar{g}_{L}$	2000; 0.2; 60; 0.3	mS cm ^{−2}	calibrated
NavC C-fiber	E_{Na} ; E_K ; E_{L}	+50; −77; −55	mV	–
NavC C-fiber	m_8 : $V_{1/2}$, k , τ	−25, 6, 1.5 (τ varied 0.05–1.5)	mV, mV, ms	literature range, τ voltage independent
NavC C-fiber	h_8 : $V_{1/2}$, k , τ	−42, 6, 2	mV, mV, ms	calibrated (literature $\approx$ −30 mV, 17 ms)
NavC C-fiber	m_9 : $V_{1/2}$, k , τ	−50, 5, 10	mV, mV, ms	literature range
Compartment	ρ_e ; A_e (production)	100; 16.40	Ω cm; μ m ²	A_e : earlier pair geometry ($w_{eq} = 0.50$ μ m at $n = 1$)
Compartment	κ (λ_e)	10 ⁹ (31.6 μ m); varied 10 ⁶ –10 ¹¹	m ^{−2}	phenomenological, calibrated in the earlier version
Compartment	lesion	$z = 2.5$ – 7.5 (varied 1–7 mm long)	mm	grounded bulk outside
Stimulus	A β pulse at node 0	0.2 ms, 2×1.511 nA = 3.02 nA	–	threshold of the uncoupled fiber
Numerics	L ; Δz ; Δt	10 mm; 5 μ m; 1 μ s	–	monolithic backward Euler

Table 1. Model parameters, as defined in `src/ephaptic/params.py` (this table is generated from that module). “Modelling choice”, “calibrated” and “assumption” mark values that are not taken directly from the cited source.

2.3 Extracellular compartment

The production compartment has a fixed cross-section $A_e = 16.4 \mu\text{m}^2$. This is the value used in the earlier version of this study for a fiber pair: a circle circumscribing both fibers, widened by 20 nm, minus the two fiber cross-sections. Spread uniformly around one A β fiber, $16.4 \mu\text{m}^2$ corresponds to a periaxonal gap of $0.50 \mu\text{m}$; the model is a one-dimensional mean-field description and does not resolve a 20-nm cleft as such. When n fibers share A_e , the extracellular cross-section per A β fiber is A_e/n . In the compartment equation the fiber count and the cross-section enter together: dividing the equation by n replaces $(1/r_e, G_e)$ with $(1/n r_e, G_e/n)$ at the same κ , so adding fibers to a fixed compartment acts like shrinking the compartment around one fiber. The correspondence is not exact, because the C-fiber’s own axial conductance g_2 is not rescaled; it is 1.6 % of the total axial conductance at $n = 1$ and 0.08 % at $n = 25$, and the resulting deviation grows with n (Section 3.5). We therefore report every multi-fiber result also in terms of the equivalent periaxonal gap w_{eq} of a uniform sleeve with area A_e/n around a $10\text{-}\mu\text{m}$ fiber ($w_{\text{eq}} = 0.50 \mu\text{m}$ at $n = 1$, 21 nm at $n = 25$, 5 nm at $n = 100$), as a readable measure of confinement rather than as an exact mapping. We also used an alternative closure in which every fiber brings its own sleeve of width w , so that A_e grows with n and the confinement is set by w alone (Section 3.5).

κ is not a measured quantity. The baseline $\kappa = 10^9 \text{ m}^{-2}$ ($\lambda_e = 31.6 \mu\text{m}$) is the value of the earlier version, where it was chosen so that a single-pair response was a few millivolts; we disclose this calibration and vary κ over five decades ($10^6\text{--}10^{11} \text{ m}^{-2}$, $\lambda_e = 1 \text{ mm}$ to $3 \mu\text{m}$), which brackets a nearly sealed compartment and one that is short-circuited to the bulk within a few micrometres.

2.4 Numerical method

The three fields are discretized with second-order central differences ($\Delta z = 5 \mu\text{m}$, 2000 compartments) and advanced together by backward Euler ($\Delta t = 1 \mu\text{s}$): at each step the linear system for $[v_1, v_2, u_e]$ at the new time level is solved in one banded LU solve (LAPACK `gbtrf/gbtrs`, factorized once per run), so the extracellular potential is never lagged behind the membrane potentials. Gating variables are advanced by a Rush–Larsen step at the old potential, and ionic currents are evaluated at the updated gates and old potentials. A conductance-implicit variant (ionic conductances in the system matrix, refactorized every step) was used as an independent check. It agreed with the production scheme for $n \leq 50$, but at $\Delta t = 1 \mu\text{s}$ it misplaced the A β conduction-block boundary at $n = 100$ (3.6 mV vs 27.4 mV at $\Delta t = 0.1 \mu\text{s}$, which the explicit scheme reproduces at $1 \mu\text{s}$: 26.9 mV). It is therefore not used for the A β fiber. Initial conditions are the exact steady state of the coupled system, obtained by Newton’s method from the space-clamped resting potential of each membrane; it is uniform to within 10^{-6} V (resting Na^+ conductance holds the A β nodes a fraction of a microvolt above the internodes, which drives a standing extracellular potential of the same order).

The earlier version of this study computed u_e from the membrane potentials of the previous step and fed $\partial^2 u_e / \partial z^2$ back explicitly. Because the explicit term cancels a fraction $n g_1 / \Sigma_G$ of the implicit axial term, with $\Sigma_G = 1/r_e + n g_1 + g_2$, the resulting time-step error grows with n . We reproduce that scheme in the code (`run_lagged`) only to document its error (Section 3.1).

2.5 Multi-fiber volleys and temporal dispersion

Synchronous volleys use one representative A β cable whose axial conductance enters the compartment equation with weight n . Dispersed volleys use K explicit A β cables, each representing n/K fibers, all sharing the same compartment, with stimulus onsets spaced uniformly over a window of width W . Because the depolarization produced by one fiber lasts only ~ 0.1 ms, K must be large enough that the spacing $W/(K - 1)$ is well below that, or the ensemble peak merely tracks the response of one group. We therefore set K so that the spacing is at most 25 μ s ($K = 61$ at $W = 1.5$ ms), and verified convergence in K and in Δz (Table S5). Because the cost of a step grows with K times the number of compartments, the dispersion sweep uses $\Delta z = 20$ μ m; the grid dependence is reported in the same table. W was varied from 0 to 2 ms.

2.6 Sensitivity analysis

Around the production configuration we varied, one or two factors at a time:

- the number of synchronous fibers n (1–300, i.e. $w_{eq} = 0.5$ μ m to 1.7 nm);
- κ (10^6 – 10^{11} m $^{-2}$);
- lesion length (1–7 mm);
- stimulus strength (1.2 – $20 \times$ threshold);
- the dispersion window W (0–2 ms);
- repetitive trains of 5 pulses at 50–400 Hz;
- a uniform depolarizing bias current on the C-fiber, up to the loss of stability of its resting state (HH, at 0.1 A m $^{-2}$) or to 3.5 A m $^{-2}$ (NavC, whose resting state stays stable over that range), with the equilibria and their linear stability computed from the space-clamped membrane equations, and a no-stimulus control for every bias;
- the speed of C-fiber kinetics.

For the kinetics, we used HH at 6.3–25 $^{\circ}$ C; a “time-compressed” HH membrane in which all rates *and* maximal conductances are multiplied by $s = 2$ –10, which is equivalent to dividing all time scales and the membrane time constant by s while preserving excitability (HH with standard conductances fails to conduct at 30 $^{\circ}$ C); HH with the conductances alone multiplied by s ; NavC with $\tau_{m8} = 1.5$ –0.05 ms; and a time-compressed NavC membrane (all gate time constants divided by s , all conductances multiplied by s). Every kinetic variant was first verified to conduct an AP when stimulated directly (Table 2). The variants span C-fiber conduction velocities of 0.25–1.43 m/s. Variants with scaled-up conductances have membrane time constants comparable to Δt . Their C-fiber ionic current was treated conductance-implicitly, while the A β fiber kept the explicit treatment. This agreed with $\Delta t = 0.25$ μ s runs (implicit and explicit) to within 1.3 % in subthreshold peak depolarization, with identical spike classification (e02b_fast_variants.csv).

2.7 Endpoints and threshold characterization

The primary endpoint is a **propagating C-fiber AP**: the C-fiber overshoots 0 mV over a stretch of at least 3 mm, and the overshoot front moves outward at 0.05–5 m/s, measured over the outermost 1 mm on either side. This tolerates near-simultaneous initiation at several nodes. The same criterion is applied to every run. Runs continue until the overshoot has spread over 7 mm, until the system has returned to within 0.25 mV of rest (no earlier than 4 ms, and at least 2 ms after the last stimulus), or until 25 ms. Secondary endpoints are the peak C-fiber depolarization in the lesion (ΔV , relative to the exact resting state), the peak hyperpolarization, and whether the A β AP conducts through the lesion to the last node (v_1 crossing -40 mV, with crossing times interpolated within the step).

Because no fixed voltage threshold applies to a brief, spatially non-uniform depolarization, we measured excitability directly in two ways:

1. **Strength–duration curves** of the isolated C-fiber for intracellular current pulses of 0.02–20 ms, either injected at one point or applied uniformly over the 5-mm lesion segment. We report the threshold and the peak depolarization reached by a pulse 0.5 % below threshold.
2. **An ephaptic safety factor α^*** . The $u_e(z, t)$ recorded in a closed-loop run is applied with gain α to the isolated C-fiber, and α^* is the smallest gain that evokes a propagating AP. Because the C-fiber carries at most 1.6 % of the compartment's axial conductance, removing it barely changes u_e : at $\alpha = 1$ the open-loop response reproduced the closed-loop depolarization to within 0.00 % in every case. α^* is therefore the factor by which the ephaptic drive would have to grow to excite the fiber, and $\alpha^* = 1$ means the closed-loop drive itself fires it.

3 Results

3.1 Numerical verification

The earlier lagged scheme underestimated the multi-fiber response, and its error grew with the number of fibers (Fig. 2a; Table S1). In the earlier full-length geometry and parameters, the $n = 25$ response was 6.10 mV at the earlier production step ($\Delta t = 2.5 \mu\text{s}$) and rose to 15.41 mV at $\Delta t = 0.125 \mu\text{s}$. The monolithic scheme gives 15.42 mV already at $\Delta t = 2.5 \mu\text{s}$ and 15.47 mV at $0.125 \mu\text{s}$. At $n = 50$ the two schemes differ by more than a factor of three at $\Delta t = 2.5 \mu\text{s}$ (6.28 vs 21.44 mV). Over $\Delta t = 5\text{--}0.125 \mu\text{s}$ the monolithic responses change by at most 2.4 % for $n = 1\text{--}50$. The apparent saturation of the earlier version was therefore the shape of the time-step error.

In the production configuration (Fig. 2b, c; Table S2):

- Refining Δt from $1 \mu\text{s}$ to $0.125 \mu\text{s}$ changed the peak C-fiber depolarization by at most 0.4 %, and refining Δz from $5 \mu\text{m}$ to $1.25 \mu\text{m}$ by at most 0.5 %, for both C-fiber models and $n = 1\text{--}50$.
- The conductance-implicit variant agreed with the production scheme to within 1.1 %.
- The physical current-balance residual of the compartment was at most $1\text{e-}10$ of the peak membrane source current.
- Spike classification did not change under any refinement.

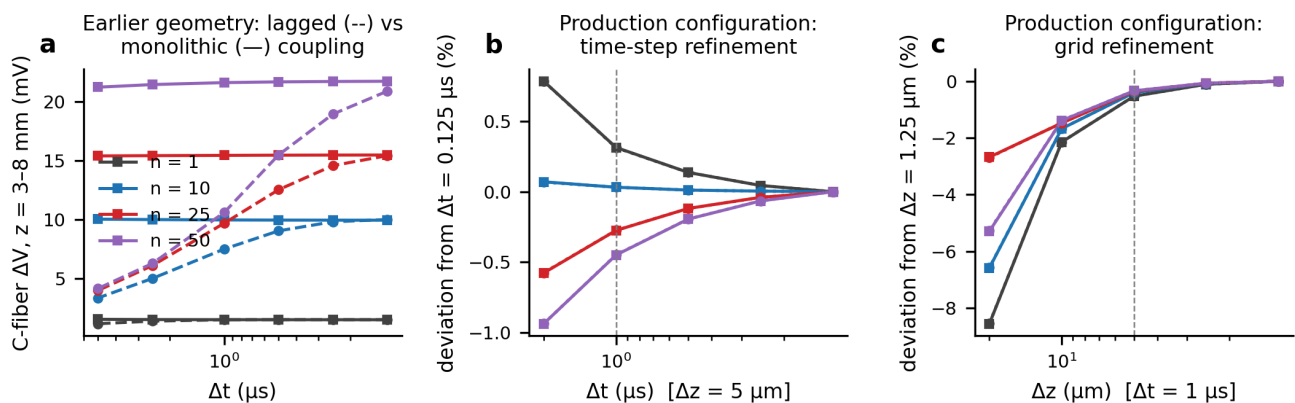


Fig. 2 Numerical verification. (a) Earlier full-length geometry and parameters: C-fiber depolarization vs time step for the lagged coupling scheme of the earlier version (dashed) and the monolithic scheme (solid); the lagged error grows with n . (b, c) Production configuration: deviation of the peak C-fiber depolarization from the finest time step (b) and grid (c) for $n = 1, 10, 25, 50$ (solid: NavC; dotted: HH); dashed lines mark the production values.

3.2 Uncoupled positive controls

The isolated A β fiber had a stimulation threshold of 1.51 nA (0.2 ms pulse at node 0). With the 2 $\times$ -threshold stimulus used throughout (3.02 nA), it conducted saltatorily at 40.4 m/s with an AP excursion of 80.8 mV at $z = 4$ mm (Fig. 3a); node 0 peaked at +14.4 mV. For comparison, the 100-nA stimulus of the earlier version drove node 0 of the uncoupled fiber to +786 mV. Directly stimulated, the HH C-fiber rested at -65.00 mV and conducted at 0.45 m/s with an AP peak of 38.0 mV (Fig. 3b). The NavC C-fiber rested at -66.82 mV and conducted at 0.25 m/s, with an AP peak of 37.7 mV and APD₅₀ = 7.3 ms (Fig. 3c). Every kinetic variant used in Section 3.7 conducted, with conduction velocities of 0.25–1.43 m/s (Table 2). The exception is HH with standard conductances at ≥ 30 °C, which failed to conduct and was excluded.

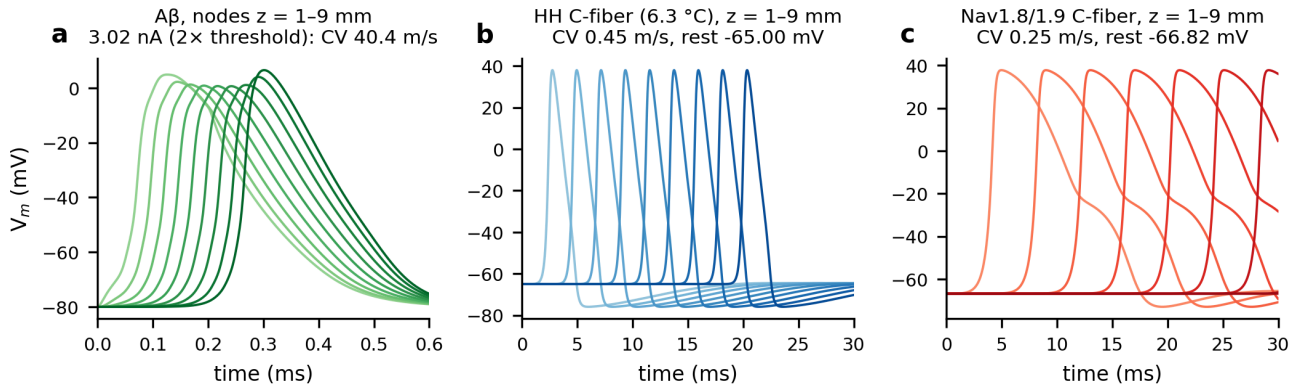


Fig. 3 Uncoupled positive controls. (a) A β action potential at nodes $z = 1-9$ mm (2 $\times$ threshold stimulus at node 0). (b) HH and (c) NavC C-fiber action potentials at $z = 1-9$ mm after direct stimulation at $z = 0$.

C-fiber membrane	rest (mV)	τ_m at rest (ms)	AP peak (mV)	CV (m/s)	APD₅₀ (ms)	conducts
HH, 6.3 °C (Phase 1)	-65.00	1.48	38.0	0.46	1.58	yes
HH, 10 °C (φ = 1.5)	-65.00	1.48	35.5	0.52	1.08	yes
HH, 15 °C (φ = 2.6)	-65.00	1.48	30.5	0.62	0.67	yes
HH, 20 °C (φ = 4.5)	-65.00	1.48	23.1	0.72	0.42	yes
HH, 25 °C (φ = 7.8)	-65.00	1.48	12.3	0.81	0.29	yes
HH, 30 °C (φ = 14)	-65.00	1.48	-3.2	–	0.22	no
HH, rates & conductances $\times 2$	-65.00	0.74	37.9	0.64	0.78	yes
HH, rates & conductances $\times 3$	-65.00	0.49	37.9	0.79	0.52	yes
HH, rates & conductances $\times 4$	-65.00	0.37	37.9	0.91	0.39	yes
HH, rates & conductances $\times 5$	-65.00	0.30	37.9	1.01	0.31	yes
HH, rates & conductances $\times 7$	-65.00	0.21	37.7	1.20	0.22	yes
HH, rates & conductances $\times 10$	-65.00	0.15	37.8	1.43	0.15	yes
HH, conductances $\times 2$	-65.00	0.74	40.7	0.49	1.54	yes
HH, conductances $\times 5$	-65.00	0.30	42.3	0.53	1.52	yes
HH, conductances $\times 10$	-65.00	0.15	42.9	0.52	1.52	yes
NavC, $\tau_{m8} = 1.5$ ms (Phase 2)	-66.82	1.37	37.7	0.25	7.31	yes
NavC, $\tau_{m8} = 1$ ms	-66.82	1.37	43.9	0.34	7.17	yes
NavC, $\tau_{m8} = 0.5$ ms	-66.82	1.37	48.1	0.50	7.07	yes
NavC, $\tau_{m8} = 0.2$ ms	-66.82	1.37	49.5	0.73	7.02	yes
NavC, $\tau_{m8} = 0.1$ ms	-66.82	1.37	49.8	0.96	6.99	yes
NavC, $\tau_{m8} = 0.05$ ms	-66.82	1.37	49.9	1.25	6.97	yes
NavC, time constants $\div 2$, conductances $\times 2$	-66.82	0.68	37.7	0.35	3.65	yes
NavC, time constants $\div 5$, conductances $\times 5$	-66.82	0.27	37.7	0.55	1.46	yes
NavC, time constants $\div 10$, conductances $\times 10$	-66.82	0.14	37.7	0.78	0.72	yes

Table 2. Uncoupled positive controls of the C-fiber membranes (1-ms pulse at $z = 0$; conduction velocity between $z = 3$ and 7 mm; APD_{50} and AP peak at $z = 5$ mm). τ_m at rest = C_m/g_{total} at the resting potential. “Rates & conductances $\times s$ ” is the time-compressed HH membrane of Section 2.6.

3.3 Verification of the A β source waveform and of the C-fiber passive properties

Because the A β action potential is the source of everything that follows, its gating was checked directly (Table S7; Fig. S3). In the CRRSS membrane used here, steady-state inactivation falls with depolarization, from $h_\infty = 0.750$ at -80 mV to 0.289 at -70 mV and 0.001 at -40 mV (half-inactivation at -74 mV), while activation rises from $m_\infty = 0.003$ to 0.980 over the same range; $\tau_m = 12$ μs and $\tau_h = 192$ μs at rest. The action potential at a node is therefore a transient: it rises from -79.99 mV to $+0.7$ mV and returns to within 1 mV of rest 0.65 ms after the stimulus, ending the 5 -ms run at -79.99 mV. A paired-pulse protocol gives the expected refractory behaviour: a second stimulus 0.50 ms after the first evokes no propagating action potential, recovery begins at 0.75 ms, and by 2.0 ms the second action potential matches the first in peak and velocity (Table S7).

The C-fiber was characterized passively under the same geometry (Table S8). Measured input resistance at the injection site agrees with the small-signal cable prediction $0.5 \cdot \sqrt{(r_m r_i)}$ to within 0.2% (69 M Ω for HH, 61 M Ω for NavC), and the rheobase of a 1 -ms point injection is 0.16 nA and 0.70 nA respectively. The small-signal length constants are 198 and 176 μm ; they are shorter than the values obtained from the resting chord conductance (260 and 250 μm) because the slope conductance at rest is about twice the chord conductance.

3.4 Anatomy of the ephaptic drive

When the A β AP passes through the lesion, each active node draws current from the shared compartment, and u_e develops a negative trough at the node flanked by positive lobes (Fig. 4a). The C-fiber’s intracellular potential barely changes, so its transmembrane potential mirrors $-u_e$. It is depolarized under the active A β node and hyperpolarized on either side (Fig. 4b, c). The depolarization is brief and narrow. For $n = 10$ its full width at half maximum is 70 μs in time (46 μs for $n = 1$, 140 μs for $n = 25$) and about 1010 μm in space (Fig. 4d, e). This follows from the form of the drive. The ephaptic input to the C-fiber is the activating function $g_2 \partial^2 u_e / \partial z^2$ (Rattay, 1986). Its integral along the fiber is $g_2 [\partial u_e / \partial z]$ evaluated at the ends, which vanishes because u_e is zero outside the lesion: the drive injects no net charge into the C-fiber, it only redistributes charge from the flanks to the site under the active node.

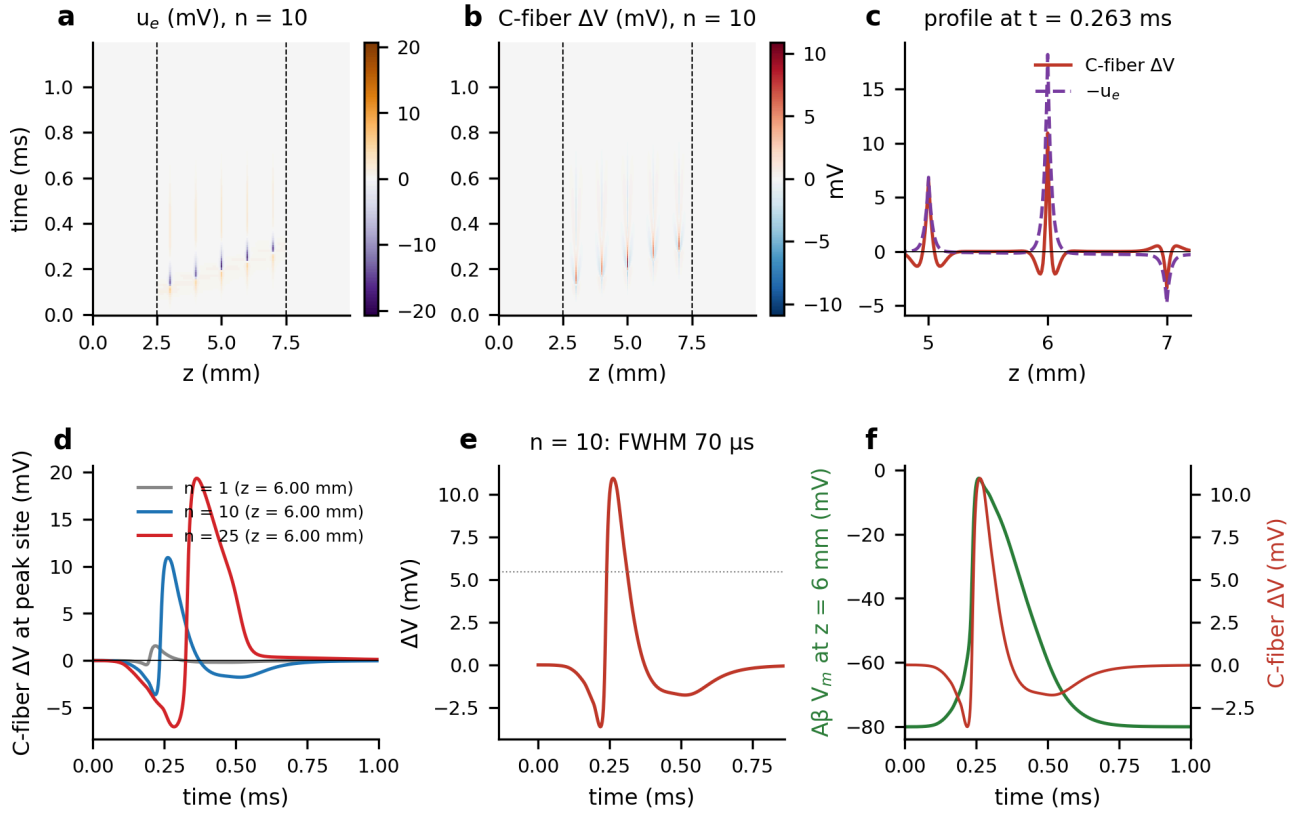


Fig. 4 The ephaptic drive (NavC, $n = 10$, production configuration). (a) Extracellular potential $u_e(z, t)$; dashed lines mark the lesion. (b) C-fiber depolarization $\Delta V(z, t)$. (c) Profiles of ΔV and $-u_e$ at the time of the peak. (d) ΔV at the peak site for $n = 1, 10, 25$. (e) Time course at the peak site with its half-maximum level. (f) A β membrane potential at the 6-mm node and the C-fiber depolarization at the same position.

3.5 Synchronous volleys and the extracellular cross-section per fiber

A single A β AP depolarized the C-fiber by 1.56 mV (NavC; HH 1.56 mV; Fig. 5a; Table 3). The response grew with the number of synchronous fibers: 10.9 mV at $n = 10$, 19.3 mV at $n = 25$ and 23.5 mV at $n = 50$. It peaked at 26.9 mV for $n = 100$, where the lesion peak of the C-fiber membrane was -39.9 mV. HH and NavC responses differed by at most 0.03 mV, as expected for a membrane that responds passively. Hyperpolarization of the flanks grew in parallel (-9.5 mV at $n = 25$).

The response does not saturate. It is limited by the source. Sharing a fixed compartment among more fibers loads the A β axons themselves: the A β conduction velocity inside the lesion fell from 38.4 m/s at $n = 1$ to 16.0 m/s at $n = 22$. From $n = 25$ (equivalent gap $w_{eq} \approx 21$ nm) the A β AP failed inside the lesion (Fig. 5b). The depolarization was then generated only by the nodes reached before failure. Above $n \approx 100$ the A β AP was blocked at the first lesion node and the C-fiber response collapsed to 4.1 mV. No C-fiber AP occurred at any n from 1 to 300 in either model.

n	w_eq (nm)	ΔV NavC (mV)	ΔV HH (mV)	peak hyperpol. (mV)	u_e min (mV)	A β through lesion	A β CV in lesion (m/s)	α^* NavC	α^* HH	C-fiber AP
1	497	1.56	1.56	-0.5	-3.1	yes	38.4	43.2	42.0	no
2	255	2.97	2.97	-0.9	-5.9	yes	36.7	–	–	no
5	103	6.53	6.53	-2.2	-12.8	yes	32.5	–	–	no
10	52	10.93	10.93	-4.3	-21.0	yes	26.9	6.4	6.2	no
15	35	14.23	14.23	-6.2	-27.0	yes	22.3	–	–	no
20	26	16.99	16.98	-7.9	-31.6	yes	17.9	–	–	no
25	21	19.35	19.34	-9.5	-35.1	no	–	4.6	4.4	no
30	17	20.32	20.31	-11.0	-37.0	no	–	–	–	no
50	10	23.53	23.51	-15.8	-42.6	no	–	50.2	3.7	no
100	5	26.93	26.91	-23.7	-49.0	no	–	20.1	3.1	no
150	3	4.14	4.16	-27.5	-2.2	no	–	–	–	no
300	2	4.67	4.68	-31.1	-4.0	no	–	–	–	no

Table 3. Synchronous volleys in the production configuration (5-mm lesion, $A_e = 16.4 \mu\text{m}^2$, $\kappa = 10^9 \text{ m}^{-2}$, $2\times$ -threshold A β stimulus). w_{eq} : equivalent periaxonal gap of the cross-section per A β fiber, A_e/n . ΔV : peak C-fiber depolarization in the lesion relative to the exact resting state. α^* : ephaptic safety factor, the gain on the recorded u_e needed to evoke a propagating C-fiber AP (Section 3.6).

Adding fibers to a fixed compartment acts like shrinking the compartment around one fiber: one A β fiber in a cross-section A_e/n gave responses that differed from the n -fiber result by 0.9 % at $n = 2$, rising to 15 % at $n = 50$ as the C-fiber's own unscaled axial conductance becomes relatively more important (e04_mean_field.csv). These results are therefore mainly a statement about the extracellular cross-section per active fiber. We checked this with a closure in which every fiber brings its own sleeve of width w (Fig. 5c). The response is then independent of n to within 15 % whenever the A β AP conducts. It is set by w alone: 0.77 mV at 1 μm , 6.7 mV at 100 nm and 19.9 mV at 20 nm. It coincides with the shared-compartment results plotted at $w_{eq}(n)$, apart from the small extracellular area that the C-fiber's own sleeve adds. With per-fiber sleeves, A β conduction failed below $w = 30$ nm, so the largest depolarization reached with a conducting A β source was 15.6 mV, at a sleeve 30 nm wide around every active fiber sustained over millimetres. The same confinement that strengthens the drive blocks the A β source.

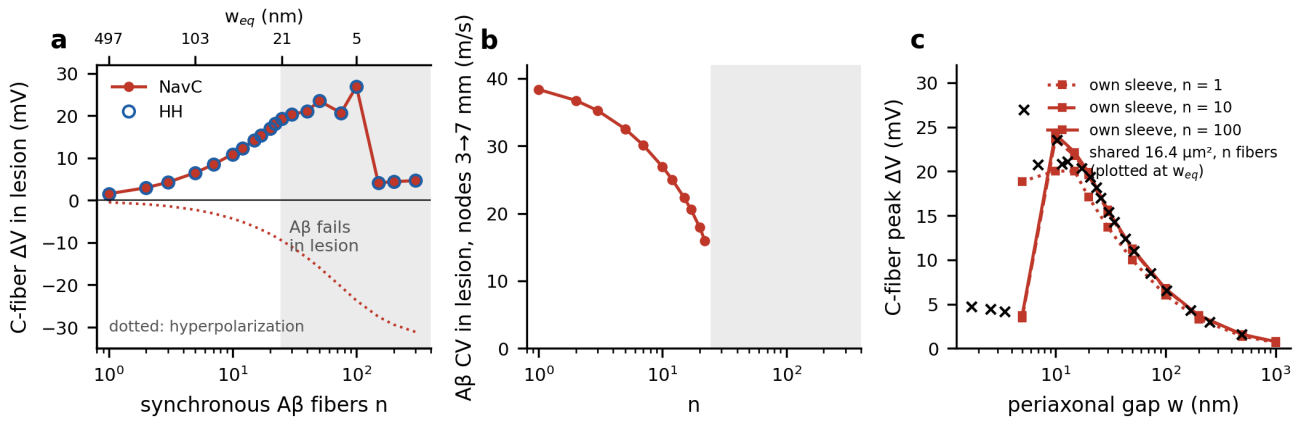


Fig. 5 Synchronous volleys. (a) Peak C-fiber depolarization (solid) and hyperpolarization (dotted) in the lesion vs the number of synchronous A β fibers n sharing the $16.4\text{-}\mu\text{m}^2$ compartment; numbers above give the equivalent periaxonal gap w_{eq} ; shading: A β conduction fails inside the lesion. (b) A β conduction velocity between the 3- and 7-mm nodes. (c) Per-fiber-sleeve closure: response vs sleeve width for $n = 1, 10, 100$ (NavC), with the shared-compartment results plotted at $w_{eq}(n)$ (crosses).

3.6 How far is the drive from threshold?

A fixed voltage threshold does not apply to this drive. For point current injections into the isolated C-fiber (Fig. 6a, b; Table S3), the membrane potential reached by a pulse just below threshold depended strongly on the pulse duration. A 20-ms pulse fired the NavC fiber from about -31 mV, whereas a 0.1-ms pulse, the duration of the ephaptic transient, could take the injection site to 39 mV without firing (HH: -53 and -30 mV). Brief local depolarizations far above the nominal “threshold” are shunted along the cable before Na^+ channels can open enough.

The ephaptic safety factor α^* measures the margin directly (Fig. 6c; Table 3). Applying the recorded u_e to the isolated C-fiber reproduced the closed-loop response to within 0.00 % at $\alpha = 1$. For the Phase 2 membrane the drive would have to be multiplied by $\alpha^* = 43.2$ at $n = 1$, 6.4 at $n = 10$ and 4.6 at $n = 25$ to evoke an AP (HH: 42.0, 6.2 and 4.4). Across the standard membranes (HH at 6.3 and 25 °C; NavC with $\tau_{m8} = 1.5$ and 0.05 ms), the smallest safety factor at any n was 1.3. α^* does not fall monotonically with n : once the A β AP fails inside the lesion the drive collapses onto one or two nodes, and although its peak amplitude is larger, a shorter depolarized stretch is harder to turn into a propagating AP (for the Phase 2 membrane α^* rises from 4.6 at $n = 25$ to 50.2 at $n = 50$).

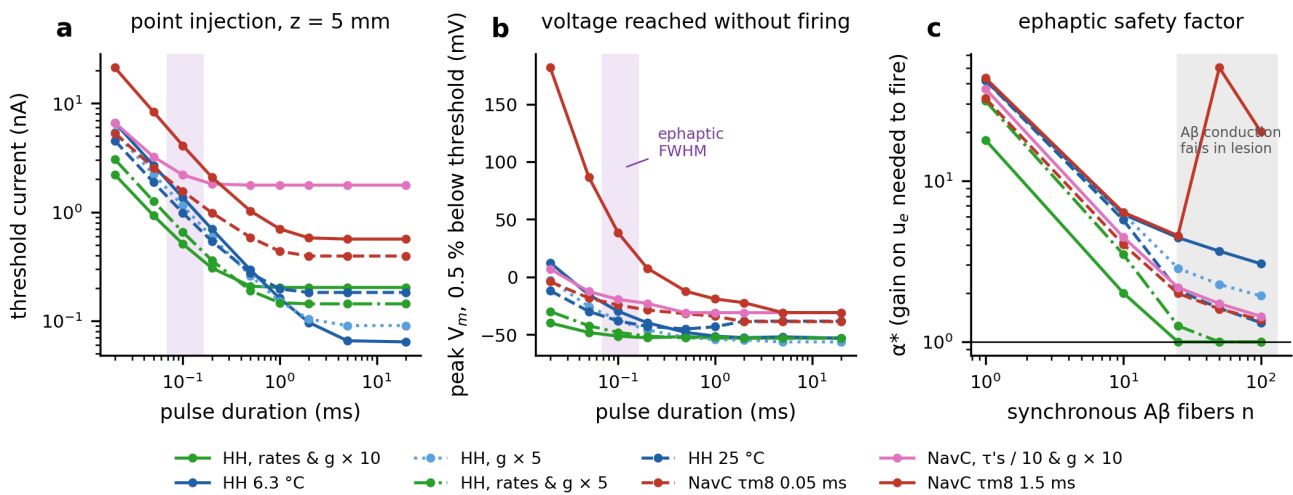


Fig. 6 Excitability of the C-fiber for brief, local inputs. (a) Strength–duration curves for point injection at $z = 5$ mm; shading marks the duration of the ephaptic depolarization. (b) Membrane potential at the injection site for a pulse 0.5 % below threshold. (c) Ephaptic safety factor α^* , the gain on the recorded u_e needed to evoke a propagating C-fiber AP, vs n ; $\alpha^* = 1$ (line) would mean that the closed-loop drive itself fires the fiber.

3.7 Sensitivity analysis

Extracellular leak. Over five decades of κ ($\lambda_e = 1$ mm to 3 μm) and $n = 1$ –200, none of the 154 runs with the standard membranes produced a C-fiber AP or even an overshoot of 0 mV (Fig. 7a). κ shifts where the source fails: a weakly leaky compartment (small κ) blocks A β conduction at smaller n , while a strongly leaky one shunts u_e . The largest depolarization was 33.9 mV ($\kappa = 3 \times 10^{10} \text{ m}^{-2}$, $n = 200$). The single-pair response ranged from 0.24 to 2.18 mV.

C-fiber kinetics. Speeding up gating alone had almost no effect (Fig. 7b). Reducing τ_{m8} from 1.5 to 0.05 ms, or warming the HH membrane from 6.3 to 25 °C (ϕ up to 7.8), changed the peak response by at most 0.1 % and 1.5 %, respectively, and never produced an AP, although these variants conduct up to five times faster (Table 2). Scaling the HH conductances alone ($\times 2$ –10) did not produce an AP either. C-fiber APs appeared only when gating rates and conductances were scaled together, that is, when the whole membrane was made faster, with its time constant at rest shortened to 0.37 ms or less:

- Time-compressed HH fired once $s \geq 4$ (membrane time constant at rest ≤ 0.37 ms), never at $s \leq 4$. At $\kappa = 10^9$ m^{-2} the smallest volley that fired was $n = 100$ at $s = 4$, 50 at $s = 5$, 25 at $s = 7$ and 25 at $s = 10$, that is, for extracellular cross-sections per fiber below ~ 20 nm equivalent gap.
- Time-compressed NavC never fired, even at $s = 10$ (membrane time constant 0.14 ms), because Nav1.8 activation is half-maximal at -25 mV, some 42 mV above rest, so speed alone does not bring it into range.

The APs were initiated at A β nodes inside the lesion ($z = 3.0$ – 6.0 mm) and propagated along the whole cable at 0.9–1.4 m/s. In these variants α^* fell to 1.0 (HH $\times 10$).

Sensitizing bias. A uniform depolarizing bias moves the C-fiber towards threshold (Fig. 7c; Table S4). The HH resting state lost stability between 0.095 and 0.1 A m^{-2} (rest -59.8 mV), beyond which the fiber fires spontaneously; within the stable range ephaptic input evoked no AP, the highest C-fiber peak being -33.9 mV. The NavC resting state stayed stable throughout the tested range (to 4 A m^{-2} , rest -40.8 mV), acquiring a second, depolarized stable state above 2.75 A m^{-2} without losing the resting one. Even at 3.5 A m^{-2} with $n = 100$, where the C-fiber in the lesion reached -15.9 mV, no AP was evoked. No-stimulus controls confirmed that no run fired spontaneously.

Lesion length and stimulus strength. Lesions of 1–7 mm and A β stimuli of 1.2 – $20 \times$ threshold produced no C-fiber AP (largest depolarizations 26.9 and 21.3 mV; Fig. S2). For $n \leq 10$ the stimulus strength changed the response by at most 2.9 %, because the stimulus site lies outside the lesion.

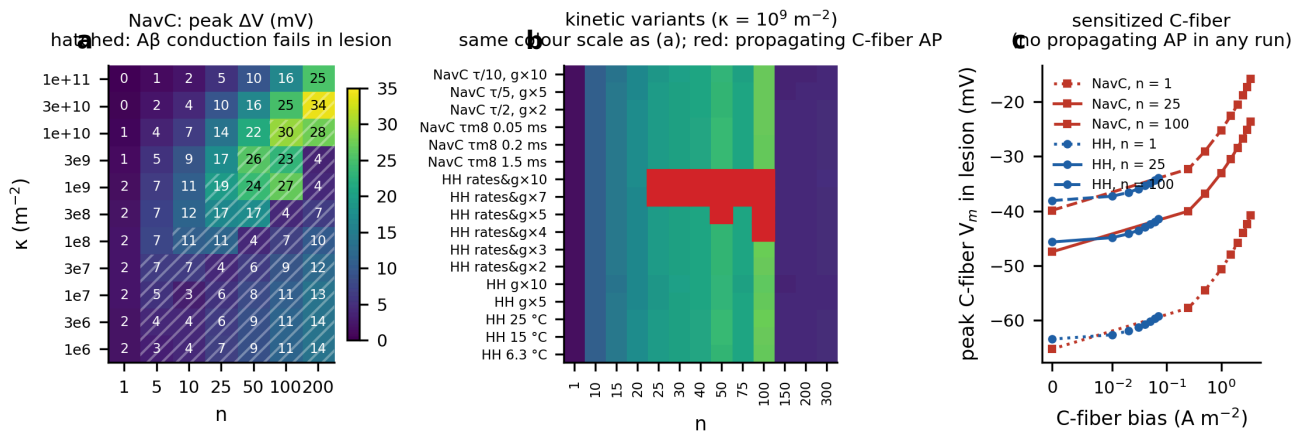


Fig. 7 Sensitivity analysis. (a) Peak C-fiber depolarization (mV, NavC) over κ and n ; hatching: A β conduction fails in the lesion; no C-fiber AP occurred. (b) Kinetic variants ($\kappa = 10^9$ m^{-2}): peak depolarization, with red cells marking propagating C-fiber APs. (c) Peak C-fiber membrane potential in the lesion vs a uniform depolarizing bias, up to the loss of resting stability (HH) or to 3.5 A m^{-2} (NavC).

3.8 Temporal dispersion and repetitive trains

Spreading the onsets of the volley over a window W attenuated the drive steeply (Fig. 8a, b), because the depolarization produced by each fiber lasts only ~ 0.1 ms. At $n = 25$ (NavC), $W = 0.2$, 0.5 and 1.5 ms reduced the peak by 56 %, 85 % and 95 %, respectively; 50 % attenuation was reached at $W \approx 0.19$ ms. The number of onset phases matters: at $W = 1.5$ ms the peak falls from 2.23 mV with $K = 11$ to 0.91 mV with $K = 61$ and 0.89 mV with $K = 81$ (1.4 % apart), because too few phases leave the groups temporally resolved; the production values use the converged spacing (Table S5). At $n = 50$ the dependence on W is not monotone at small W : the synchronous volley blocks its own A β action potentials inside the lesion, and spreading the onsets lowers the extracellular potential each fiber has to drive its current through, so the peak first rises from 22.4 to 25.6 mV at $W = 0.10$ ms, and the A β action potentials conduct again from $W = 0.2$ ms (62 m/s), before the attenuation takes over. Trains of 5 pulses at 50–400 Hz evoked no C-fiber AP. The per-pulse peak changed by at most 0.04 % over the train at ≤ 100 Hz and by at most 0.27 % at any frequency (HH, $n = 25$, 400 Hz; Fig. 8c), so there was no pulse-to-pulse build-up.

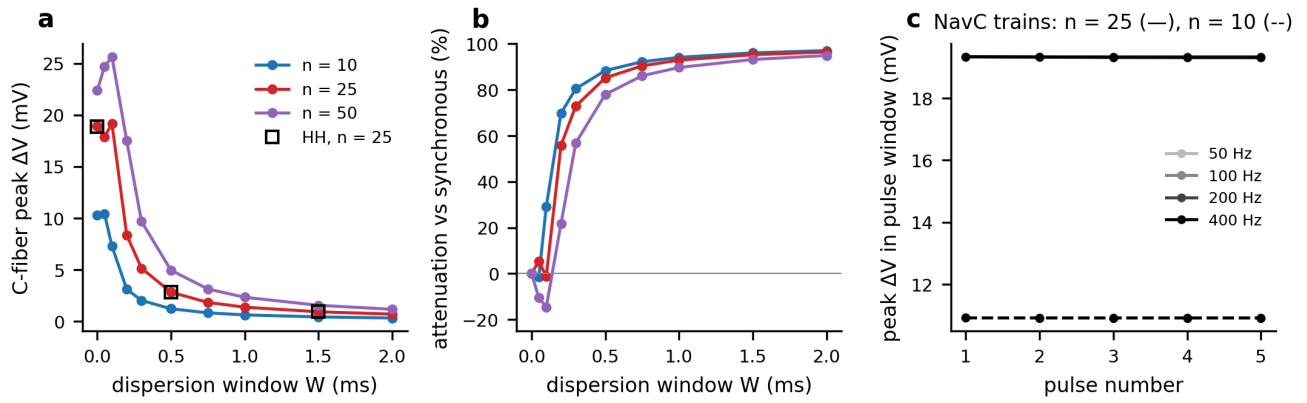


Fig. 8 Temporal dispersion and trains. (a) Peak C-fiber depolarization vs the width W of the onset-dispersion window (NavC; open squares: HH check at $n = 25$). (b) Attenuation relative to the synchronous volley. (c) Per-pulse peak depolarization in 5-pulse trains at 50–400 Hz (NavC).

3.9 The earlier full-length geometry

To relate these results to the earlier version of this study, we repeated the synchronous sweep with the shared compartment spanning the whole cable, so that the A β electrode lies inside it (Fig. S1; Table S6). Two treatments of the electrode current were compared: returning it through the compartment, as the production model does, and omitting it from the extracellular source, as the earlier version did.

With the 2 $\times$ -threshold stimulus, no propagating C-fiber AP occurred for $n \leq 50$ under either treatment, and the interior depolarizations matched the focal-lesion results (19.3 mV at most). With the earlier 100-nA stimulus the electrode site became non-physiological under both treatments, and C-fiber APs appeared there. Omitting the electrode current from the extracellular source, as the earlier version did, a propagating AP was launched at the stimulated end ($z = 0$) from $n = 2$ with the HH membrane, conducting at 0.45 m/s; the Nav1.8/1.9 membrane overshoot 0 mV at the same site from $n = 2$, but that overshoot did not propagate. Returning the electrode current through the compartment drove u_e at the electrode to 676 mV and the A β membrane at node 0 to +556 mV at $n = 25$, and launched a propagating AP from $n = 25$ (HH) and $n = 50$ (NavC, 0.25 m/s from $z = 0.07$ mm). The weakest electrode current that did this was 10 nA (HH, $n = 25$, omitted), 3 $\times$ the physiological stimulus used here. This is the spike that the earlier version recorded during its 100-Hz train. Either way, the phenomenon belongs to a suprathreshold electrode placed inside the extracellular compartment, not to ephaptic coupling between fibers.

The same applies to the low- κ ($\kappa = 3 \times 10^7 \text{ m}^{-2}$) threshold crossings that the earlier version attributed to numerical instability. Propagating C-fiber APs occurred there only with the 100-nA electrode (100 nA, $n = 1$, omitted, $A_e = 16 \mu\text{m}^2$; 100 nA, $n = 25$, omitted, $A_e = 16 \mu\text{m}^2$; 100 nA, $n = 25$, omitted, $A_e = 53 \mu\text{m}^2$; 100 nA, $n = 25$, returned, $A_e = 16 \mu\text{m}^2$; 100 nA, $n = 25$, returned, $A_e = 53 \mu\text{m}^2$), at the C-fiber conduction velocity (0.44 m/s from $z = 0.29$ mm); with a 2 $\times$ -threshold stimulus no crossing occurred at that κ (largest interior depolarization 2.2 mV).

4 Discussion

4.1 Principal findings

With the intracellular and extracellular potentials advanced together, every endpoint converged. With a physiological A β stimulus applied outside the lesion, the model gives a consistent picture:

1. **A single A β fiber depolarizes the C-fiber by 1.56 mV.** The depolarization is brief (~ 0.1 ms) and spatially narrow, and it sits between two hyperpolarized flanks.

2. **Synchronous volleys add up without saturating.** The response is limited by the source: the more fibers share a compartment (equivalently, the smaller the extracellular cross-section per fiber), the larger the drive, until the A β action potentials themselves fail inside the lesion. The largest depolarization was 26.9 mV in the production configuration and 33.9 mV over five decades of κ .
3. **These brief, local depolarizations are far from threshold.** A fixed voltage threshold such as -35 mV does not apply. For the Phase 2 membrane the drive would have to grow 4.6-fold at $n = 25$ to evoke an action potential, and the safety factor was at least 1.3 for all standard membranes at any n .
4. **No C-fiber action potential occurred with either standard membrane.** This held over the number of fibers, the extracellular leak, lesion length, stimulus strength, repetitive trains and sensitizing bias currents. A 0.5-ms onset dispersion removed 85 % of the drive.
5. **Excitation required a fast membrane that activates close to rest, in a very tight compartment.** Action potentials appeared only when the HH membrane was made faster as a whole (rates and conductances scaled together, time constant at rest ≤ 0.37 ms) and the extracellular cross-section per fiber was small (w_{eq} of a few tens of nanometres). The Nav1.8/Nav1.9 membrane never fired, at any speed.

4.2 Why faster gating does not help, and what does

We expected the slow C-fiber activation ($\tau_{m8} = 1.5$ ms; HH at 6.3 °C) to be what prevents firing. It is not. Reducing τ_{m8} thirty-fold or warming the HH membrane to 25 °C left the response unchanged within 0.1 %. The reason is the shape of the drive. The ephaptic input to the C-fiber is the activating function $g_2 \partial^2 u_e / \partial z^2$ (Rattay, 1986), which deposits charge in a region of tens of micrometres under the active A β node and removes it from the flanks; because u_e vanishes outside the lesion, its integral along the fiber is exactly zero. The C-fiber's passive length constant at rest (198 μ m for HH, 176 μ m for NavC) is several times longer than this region. The depolarized patch is therefore shunted electrotonically by the surrounding membrane and axoplasm within a fraction of the membrane time constant, before Na^+ channels can open in enough membrane to carry regenerative current. The strength–duration analysis shows the same thing from the other side: a 0.1-ms point injection could take the membrane to 39 mV without firing the NavC fiber.

Firing appeared only when the membrane time constant at rest was reduced together with the gating time constants (time-compressed HH $\times 4$ and above, τ_m at rest ≤ 0.37 ms). A depolarized patch then charges and opens Na^+ channels before it is shunted. Scaling the conductances alone, or the rates alone, was not sufficient. Speed is not sufficient either: the time-compressed Nav1.8/1.9 membrane never fired, because its activation is half-maximal at -25 mV, about 42 mV above rest, so the drive does not reach the voltage range where the current becomes regenerative however fast the membrane is. Two properties therefore have to coincide: a membrane fast enough to respond within ~ 0.1 ms, and an activation range close enough to rest. Neither is well constrained for human nociceptor axons at body temperature. The time-compressed HH variants that fired conduct at 0.9 – 1.43 m/s, within the range reported for mammalian C-fibers, whereas the NavC variants, whose activation range is closer to that of nociceptor TTX-resistant currents, did not fire at any speed tested. A test of the present conclusion with detailed, temperature-appropriate C-fiber models (Tigerholm et al., 2014; Sundt et al., 2015; Pelot et al., 2021) is therefore the most important next step.

4.3 What limits multi-fiber summation

In the earlier version, a plateau of the multi-fiber response was attributed to shunting by the parallel axoplasmic conductance of the added fibers. That plateau was the time-step error of a lagged coupling scheme, whose magnitude grows with the fraction $n g_1 / \Sigma_G$ of axial conductance carried by the A β fibers. With converged numerics the response keeps growing until the source fails. Two physical effects set the limit, and they are two sides of one mechanism.

Packing more active fibers into a fixed extracellular cross-section raises the longitudinal resistance that each fiber's return current must cross: this is what strengthens the coupling, and it is also what slows the A β action potential and finally blocks it inside the lesion. Because the equivalent formulation is a single fiber in a cross-section A_e/n , the relevant variable is the extracellular area per synchronously active fiber, not the fiber count as such. In a per-fiber-sleeve closure the response did not depend on n at all. It was set by the sleeve width: the largest depolarization reached with the A β action potential still conducting was 15.6 mV, at a sleeve 30 nm wide around every active fiber over millimetres, and tighter sleeves blocked the A β source.

4.4 Synchrony and repetition

The ephaptic transient produced by each A β fiber lasts ~ 0.1 ms, so summation across fibers requires sub-millisecond synchrony. A dispersion window of 0.5 ms removed 85 % of the drive at $n = 25$, and 1.5 ms removed 95 %. Natural tactile volleys arriving over millimetres of nerve after conduction over centimetres are unlikely to be that synchronous. Repetitive trains did not build up the response, because the C-fiber returns to rest between pulses and the extracellular compartment has no memory on the millisecond scale. The delayed spike that the earlier version recorded during a 100-Hz train was launched at a 100-nA electrode inside the compartment (Section 3.9).

4.5 Limitations

- **One-dimensional, mean-field compartment.** The model averages the extracellular potential over the cross-section. It does not resolve the geometry of real contacts between fibers, the tortuosity of the endoneurium, or fiber-to-fiber variability. A 1-D model cannot represent a 20-nm cleft between two specific fibers; it can represent only an average extracellular area per fiber. EMI models (Jäger & Tveito, 2026) or resistor-network bundle models (Capllonch-Juan & Sepulveda, 2020) are needed to test local hot spots.
- **κ is phenomenological.** We bracketed it over five decades rather than deriving it, and the conclusions held over that range.
- **No demyelination.** The A β fibers keep intact myelin in the lesion. Demyelination exposes internodal membrane, changes the source current distribution and slows conduction, and could strengthen coupling. It was not modelled.
- **Simplified C-fiber membranes.** HH is a squid model, and the NavC membrane is phenomenological, with voltage-independent time constants and calibrated inactivation. Neither includes Nav1.7 or the K⁺ subtypes of nociceptor axons. As Section 4.2 shows, the conclusions depend on the speed of the C-fiber membrane and on how close its activation range lies to rest. The geometry assumed here is pathological, not normal. A normal Remak bundle is a group of unmyelinated axons held by a non-myelinating Schwann cell, usually with Schwann-cell processes between the axons, and a large myelinated A β fiber is not ordinarily part of it (Murinson & Griffin, 2004); in intact nerve, Schwann-cell cytoplasm and basal lamina separate an A β internode from neighbouring C-fibers. The configuration modelled here therefore presupposes a specific pathological change: loss of myelin over a stretch of an A β fiber together with retraction or loss of the intervening Schwann-cell processes, so that an exposed A β axolemma and a C-fiber axolemma come to share one restricted extracellular space over millimetres. That is the configuration in which ephaptic cross-excitation has been reported experimentally in damaged nerve and in dysmyelinated roots (Rasminsky, 1980; Seltzer & Devor, 1979). Our results say what such a configuration would have to achieve electrically, not how often it occurs.
- **Idealized volleys and a single test fiber.** Fibers are identical and synchronous or uniformly dispersed. The test C-fiber is a single fiber, and Remak bundling of C-fibers (Murinson & Griffin, 2004) is not represented.

4.6 Implications

Within these limits, the model argues against direct ephaptic excitation of C-fibers by A β volleys as a stand-alone mechanism of dynamic mechanical allodynia. The coupling is real and grows with confinement and synchrony, but the conditions under which it reaches threshold are narrow. They require very tight extracellular confinement sustained over millimetres, sub-millisecond synchrony and a C-fiber membrane that is both fast and activated close to rest, and the confinement that strengthens the drive also blocks the driving A β fibers. Subthreshold ephaptic depolarizations of a few to tens of millivolts could still modulate C-fiber excitability, for example by interacting with ectopic pacemaker activity at injury sites. Experimental tests would need to show ephaptic A β $\rightarrow$ C excitation at physiological stimulus strengths, with electrode artifacts and chemically mediated cross-excitation (Amir & Devor, 1996) excluded.

5 Conclusion

In a converged closed-loop core-conductor model of an A β and a C-fiber sharing a focal lesion, a single A β action potential depolarizes the C-fiber by 1.56 mV. Synchronous volleys raise this to at most 26.9 mV, before the confinement that produces the coupling blocks A β conduction. The ephaptic depolarization lasts about 70 μ s, is local and carries almost no net charge, so it is shunted along the C-fiber long before regenerative Na⁺ current develops. With classical or Nav1.8/Nav1.9 membranes, no C-fiber action potential occurred over the explored ranges of fiber number, extracellular leak, lesion length, stimulus strength, train frequency and sensitizing bias. Excitation required a combination of extreme confinement, sub-millisecond synchrony and a C-fiber membrane that is both fast and activated close to rest; the Nav1.8/Nav1.9 membrane did not fire at any speed tested. Ephaptic A β $\rightarrow$ C coupling is therefore better viewed as a subthreshold modulator than as a stand-alone trigger of nociceptor firing. The decisive open properties are the speed and the activation range of real C-fiber membranes at body temperature.

Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author declares no competing interests.

Ethics approval, consent to participate, consent for publication. Not applicable: this is a purely computational study involving no human participants, animals, or personal data.

Data availability. All simulation outputs underlying the figures and tables (`results/data/*.csv`, with the parameter set of each experiment in the accompanying `.meta.json` files) are provided as Supplementary Information and archived in the code repository. No restrictions apply.

Code availability. The model, all experiment, figure and table scripts, and the regression tests are available at <https://github.com/MinaSaied23/ephaptic-crosstalk>. `python run_all.py` regenerates every result, figure and table, and `python -m pytest` runs the tests. The code is written in Python 3.11 (RRID:SCR_008394) with NumPy 1.26.4 (RRID:SCR_008633), SciPy 1.13.0 (RRID:SCR_008058), pandas 2.2.2 (RRID:SCR_018214) and Matplotlib 3.8.4 (RRID:SCR_008624). The banded solves use the LAPACK routines `dgbtrf` and `dgbtrs` through SciPy. No biological resources were used in this study.

Use of AI tools. [Author to confirm or edit before submission.] An AI assistant (Claude, Anthropic) was used to help re-implement the numerical solver, run the re-analysis, and draft and edit the revised text. The author checked all code, results and text and takes full responsibility for the content.

Author contributions. Mina Saied Attia Rizk: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing (original draft, review and editing), visualization.

References

- Akopian, A. N., Sivilotti, L., & Wood, J. N. (1996). A tetrodotoxin-resistant voltage-gated sodium channel expressed by sensory neurons. *Nature*, 379(6562), 257–262. <https://doi.org/10.1038/379257a0>
- Amir, R., & Devor, M. (1996). Chemically mediated cross-excitation in rat dorsal root ganglia. *Journal of Neuroscience*, 16(15), 4733–4741. <https://doi.org/10.1523/JNEUROSCI.16-15-04733.1996>
- Amir, R., & Devor, M. (2000). Functional cross-excitation between afferent A- and C-neurons in dorsal root ganglia. *Neuroscience*, 95(1), 189–195. [https://doi.org/10.1016/s0306-4522\(99\)00418-4](https://doi.org/10.1016/s0306-4522(99)00418-4)
- Arvanitaki, A. (1942). Effects evoked in an axon by the activity of a contiguous one. *Journal of Neurophysiology*, 5(2), 89–108. <https://doi.org/10.1152/jn.1942.5.2.89>
- Binczak, S., Eilbeck, J. C., & Scott, A. C. (2001). Ephaptic coupling of myelinated nerve fibers. *Physica D: Nonlinear Phenomena*, 148(1–2), 159–174. [https://doi.org/10.1016/S0167-2789\(00\)00173-1](https://doi.org/10.1016/S0167-2789(00)00173-1)
- Bokil, H., Laaris, N., Blinder, K., Ennis, M., & Keller, A. (2001). Ephaptic interactions in the mammalian olfactory system. *Journal of Neuroscience*, 21(20), RC173. <https://doi.org/10.1523/JNEUROSCI.21-20-j0004.2001>
- Bolzoni, F., & Jankowska, E. (2019). Ephaptic interactions between myelinated nerve fibres of rodent peripheral nerves. *European Journal of Neuroscience*, 50(7), 3101–3107. <https://doi.org/10.1111/ejn.14439>
- Capllonch-Juan, M., & Sepulveda, F. (2020). Modelling the effects of ephaptic coupling on selectivity and response patterns during artificial stimulation of peripheral nerves. *PLoS Computational Biology*, 16(6), e1007826. <https://doi.org/10.1371/journal.pcbi.1007826>
- Chawla, A., & Morgera, S. D. (2024). On the geometry of interacting axon tracts in relation to their functional properties. *Communications in Mathematical Biology and Neuroscience*, 2024, Article 57. <https://doi.org/10.28919/cmbn/8509>
- Chawla, A., Morgera, S. D., & Snider, A. D. (2021). On axon interaction and its role in neurological networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 18(2), 790–796. <https://doi.org/10.1109/TCBB.2019.2941689>
- Chiu, S. Y., Ritchie, J. M., Rogart, R. B., & Stagg, D. (1979). A quantitative description of membrane currents in rabbit myelinated nerve. *The Journal of Physiology*, 292, 149–166. <https://doi.org/10.1113/jphysiol.1979.sp012843>
- Cummins, T. R., Dib-Hajj, S. D., Black, J. A., Akopian, A. N., Wood, J. N., & Waxman, S. G. (1999). A novel persistent tetrodotoxin-resistant sodium current in SNS-null and wild-type small primary sensory neurons. *Journal of Neuroscience*, 19(24), RC43. <https://doi.org/10.1523/JNEUROSCI.19-24-j0001.1999>
- Devor, M., & Wall, P. D. (1990). Cross-excitation in dorsal root ganglia of nerve-injured and intact rats. *Journal of Neurophysiology*, 64(6), 1733–1744. <https://doi.org/10.1152/jn.1990.64.6.1733>
- Goldwyn, J. H., & Rinzel, J. (2016). Neuronal coupling by endogenous electric fields: cable theory and applications to coincidence detector neurons in the auditory brain stem. *Journal of Neurophysiology*, 115(4), 2033–2051. <https://doi.org/10.1152/jn.00780.2015>

- Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4), 500–544.
<https://doi.org/10.1113/jphysiol.1952.sp004764>
- Jæger, K. H., & Tveito, A. (2026). Extracellular stimulation and ephaptic coupling of neurons in a fully coupled finite element-based Extracellular-Membrane-Intracellular (EMI) model. *Frontiers in Computational Neuroscience*, 20, 1755548. <https://doi.org/10.3389/fncom.2026.1755548>
- Katz, B., & Schmitt, O. H. (1940). Electric interaction between two adjacent nerve fibres. *The Journal of Physiology*, 97(4), 471–488. <https://doi.org/10.1113/jphysiol.1940.sp003823>
- Klein, A. H., Vyshnevskaya, A., Hartke, T. V., De Col, R., Mankowski, J. L., Turnquist, B., Bosmans, F., Reeh, P. W., Schmelz, M., Carr, R. W., & Ringkamp, M. (2017). Sodium channel Nav1.8 underlies TTX-resistant axonal action potential conduction in somatosensory C-fibers of distal cutaneous nerves. *Journal of Neuroscience*, 37(20), 5204–5214. <https://doi.org/10.1523/JNEUROSCI.3799-16.2017>
- Murinson, B. B., & Griffin, J. W. (2004). C-fiber structure varies with location in peripheral nerve. *Journal of Neuropathology and Experimental Neurology*, 63(3), 246–254. <https://doi.org/10.1093/jnen/63.3.246>
- Pelot, N. A., Catherall, D. C., Thio, B. J., Titus, N. D., Liang, E. D., Henriquez, C. S., & Grill, W. M. (2021). Excitation properties of computational models of unmyelinated peripheral axons. *Journal of Neurophysiology*, 125(1), 86–104. <https://doi.org/10.1152/jn.00315.2020>
- Rasminsky, M. (1980). Ephaptic transmission between single nerve fibres in the spinal nerve roots of dystrophic mice. *The Journal of Physiology*, 305(1), 151–169. <https://doi.org/10.1113/jphysiol.1980.sp013356>
- Rattay, F. (1986). Analysis of models for external stimulation of axons. *IEEE Transactions on Biomedical Engineering*, 33(10), 974–977. <https://doi.org/10.1109/TBME.1986.325670>
- Reutskiy, S., Rossoni, E., & Tirozzi, B. (2003). Conduction in bundles of demyelinated nerve fibers: computer simulation. *Biological Cybernetics*, 89(6), 439–448. <https://doi.org/10.1007/s00422-003-0430-x>
- Schmidt, H., Hahn, G., Deco, G., & Knösche, T. R. (2021). Ephaptic coupling in white matter fibre bundles modulates axonal transmission delays. *PLoS Computational Biology*, 17(2), e1007858.
<https://doi.org/10.1371/journal.pcbi.1007858>
- Schmidt, H., & Knösche, T. R. (2022). Modelling the effect of ephaptic coupling on spike propagation in peripheral nerve fibres. *Biological Cybernetics*, 116(4), 461–473. <https://doi.org/10.1007/s00422-022-00934-9>
- Seltzer, Z., & Devor, M. (1979). Ephaptic transmission in chronically damaged peripheral nerves. *Neurology*, 29(7), 1061–1064. <https://doi.org/10.1212/wnl.29.7.1061>
- Sheheitli, H., & Jirsa, V. K. (2020). A mathematical model of ephaptic interactions in neuronal fiber pathways: could there be more than transmission along the tracts? *Network Neuroscience*, 4(3), 595–610.
https://doi.org/10.1162/netn_a_00134
- Sundt, D., Gamper, N., & Jaffe, D. B. (2015). Spike propagation through the dorsal root ganglia in an unmyelinated sensory neuron: a modeling study. *Journal of Neurophysiology*, 114(6), 3140–3153.
<https://doi.org/10.1152/jn.00226.2015>
- Sweeney, J. D., Mortimer, J. T., & Durand, D. M. (1987). Modeling of mammalian myelinated nerve for functional neuromuscular stimulation. In *Proceedings of the 9th Annual Conference of the IEEE Engineering in Medicine and*

Biology Society (pp. 1577–1578).

Tigerholm, J., Petersson, M. E., Obreja, O., Lampert, A., Carr, R., Schmelz, M., & Fransén, E. (2014). Modeling activity-dependent changes of axonal spike conduction in primary afferent C-nociceptors. *Journal of Neurophysiology*, 111(9), 1721–1735. <https://doi.org/10.1152/jn.00777.2012>